Z-transform

Causality of Systems

Ex $h[n] = a^n u[n+1]$ $\forall n<0$

$h[n] \neq 0$ $\equiv$ System is not causal.

Time domain inference :

Z-domain inference :?

$h[n] = a^n u[n]$

$h[n] = 0 \quad n<0$

Causal

$H(z) = \frac{1}{1-az^{-1}}; \quad |z|>|a|$

$\Downarrow$ pole at $z=a$

① ROC is outside the outermost pole.

② $H(z)$ is rational in z

③ there is no pole at $z=\infty$

④ degree of $N(z) \leq$ degree of $D(z)$

Observations

① Rational fn. in z

② Two zeros at $z=0$

③ Two poles — $z=a$, $z=\infty$

④ ROC: $|z|>|a|$

∵ 1 pole at $z=\infty$ ROC is not strictly outside the outermost pole

⑤ degree $N(z)$ > degree $D(z)$

$$h[n] = a^n u[n+1] = \frac{1}{a}\left\{a^{n+1} u[n+1]\right\}$$

$$x[n] \rightleftharpoons X(z)$$

$$x[n+1] \rightleftharpoons z\,X(z)$$

$$H(z) = \sum_{n=-1}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n} + a^{-1} z + a^{-1} z$$

$$= \frac{1}{1-az^{-1}} + a^{-1} z$$

$$= \frac{1 + a^{-1} z - 1}{1 - az^{-1}} = \frac{a^{-1} z}{1 - az^{-1}}$$

$x[n] = a^n u[n] \rightleftharpoons \frac{1}{1-az^{-1}}$

$h[n] = \frac{1}{a} x[n+1]$

$\xrightarrow{\frac{1}{a}} \quad \xleftarrow{a}$

$$\frac{1}{a}\,\frac{z}{1-az^{-1}}$$

$$= \frac{a^{-1} z}{1 - az^{-1}}$$

$$H(z) = \frac{a^{-1} z^2}{z - a}$$

Ex

$$h[n] = \begin{cases} a^n & n \geq 0 \\ \frac{1}{(-n)!} & n < 0 \end{cases}$$

$$h[n] = h_1[n] + h_2[n]$$

where $h_1[n] = a^n u[n]$

& $h_2[n] = \begin{cases} 0 & n \geq 0 \\ \frac{1}{(-n)!} & n < 0 \end{cases}$

$H_1(z) = \frac{1}{1-az^{-1}}$

ROC: $|z| > |a|$

$$H_2(z) = \sum_{n=-1}^{-\infty} \frac{1}{(-n)!} z^{-n}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n!} z^{n} + 1 - 1$$

$$H_2(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} - 1$$

$$= e^z - 1$$

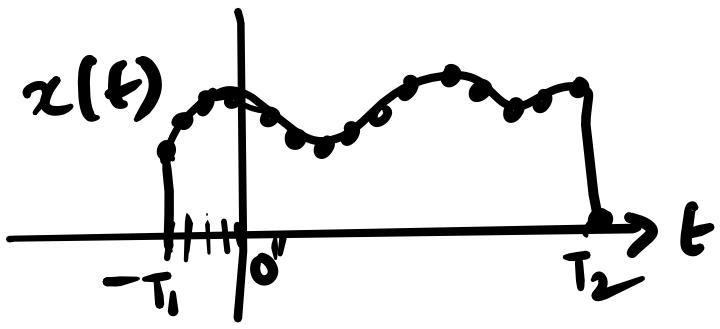
$$H(z) = \frac{1}{1-az^{-1}} + e^z - 1$$

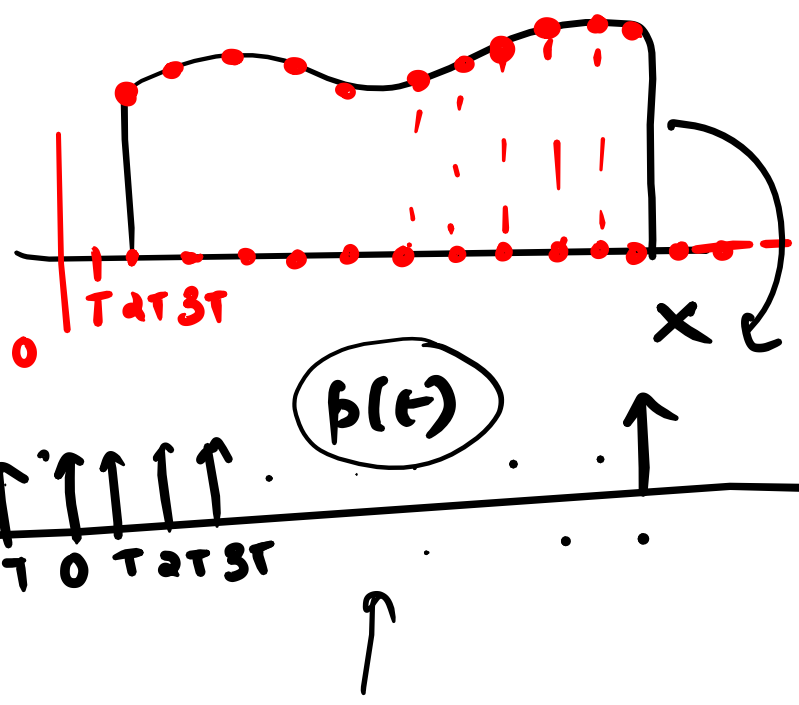
$\Downarrow$

$H(z)$ is not rational in z

$\Rightarrow$ Not a causal system

ROC is $|z| > |a|$ but should exclude $z = 0$.





Samplig

$x(t) \xrightarrow{\text{Goal}} x[n]$

$x[n] \approx x(nT)$

$T \equiv$ Samplig time

Uniform Samplig

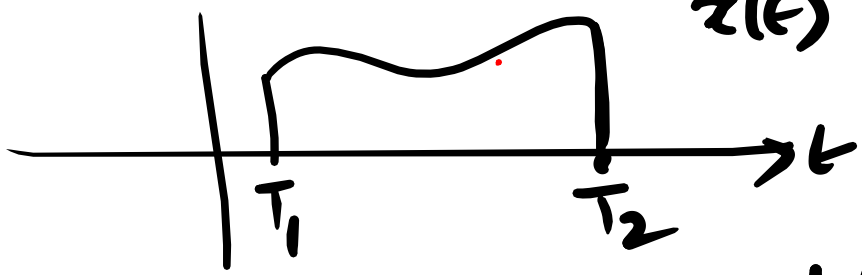
$x(t) \rightarrow \hat{x}(t)$

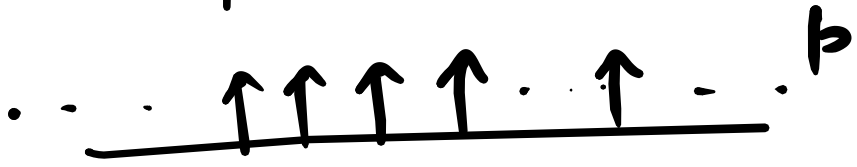
$$\rightarrow p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

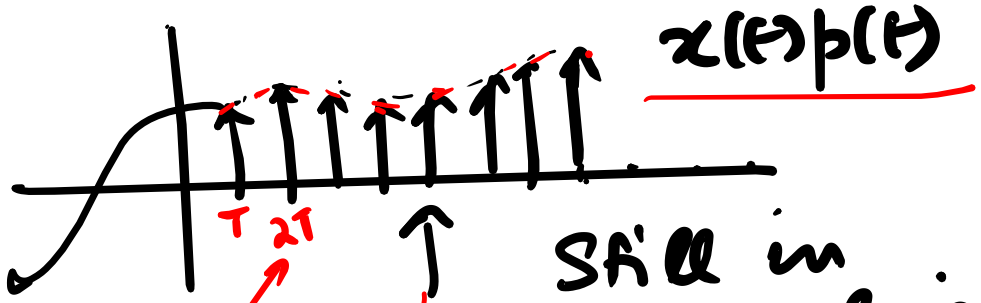
$$\hat{x}(t) = x(t)p(t)$$

$$= x(t) \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

$$\boxed{\hat{x}(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t-nT)}$$



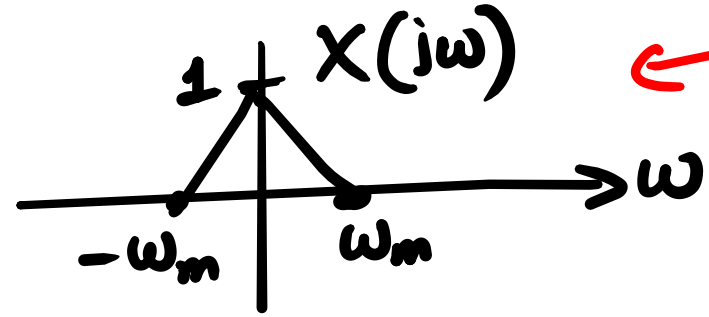




Still in continuous time domain —

$x(nT)\delta(t-nT)$

Consider $x(t)$ to be a band limited signal.



← Rich in low frequencies

$$X(j\omega) = 0 \; ; \; |\omega| > \omega_m$$

$$x(t)\,p(t) \rightleftharpoons \frac{1}{2\pi}\left[X(j\omega) * P(j\omega)\right]$$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT) \leftarrow \text{periodic}$$

$$= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad \text{F.S. representation}$$

$$p(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\frac{2\pi}{T}t}$$

ω_0

$x(t) * \delta(t-t_0) = x(t-t_0)$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\frac{2\pi}{T}t}$$

Let us assume

$\omega_m << \omega_0$

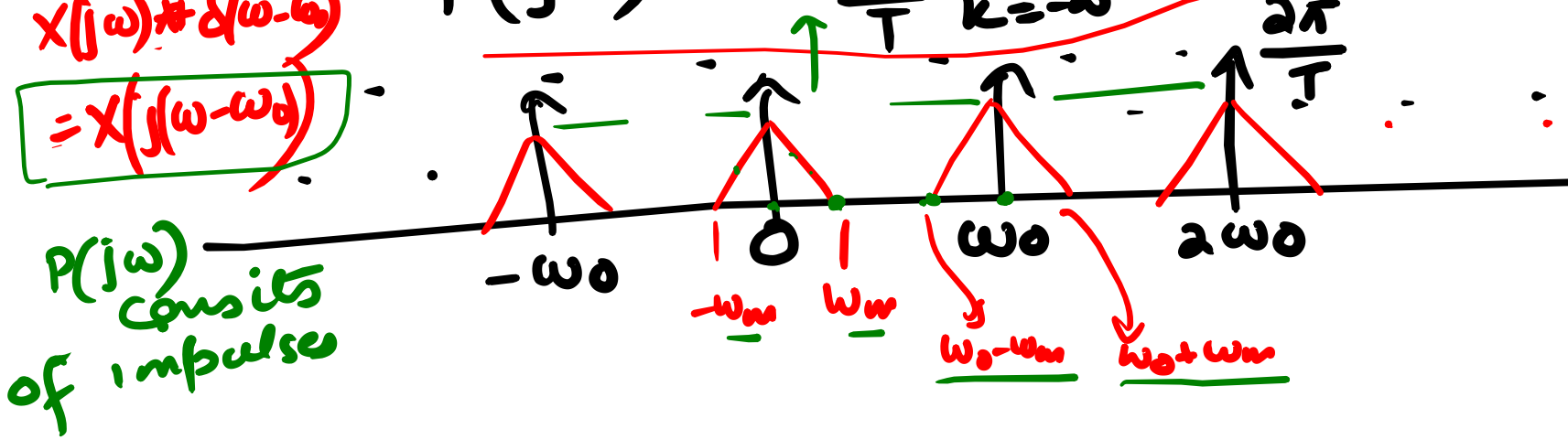
$$1 \xrightarrow{F.T} 2\pi\,\delta(\omega)$$

$$\sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\omega_0 t} \rightleftharpoons \sum_{k=-\infty}^{\infty} \frac{2\pi}{T}\delta(\omega - k\omega_0)$$

$$\frac{1}{2\pi}\{X(j\omega) * P(j\omega)\} = ?$$

$$P(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$

$X(j\omega) * \delta(\omega-\omega_0) = X(j(\omega-\omega_0))$

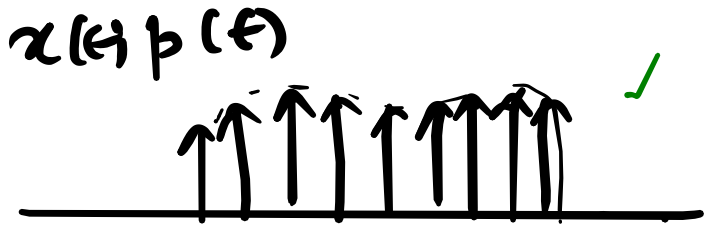


P(jw) consists of impulses

where $\omega_0 - \omega_m = \omega_m$

or $\omega_0 = 2\omega_m$

the two successive clips of spectrum touch each other

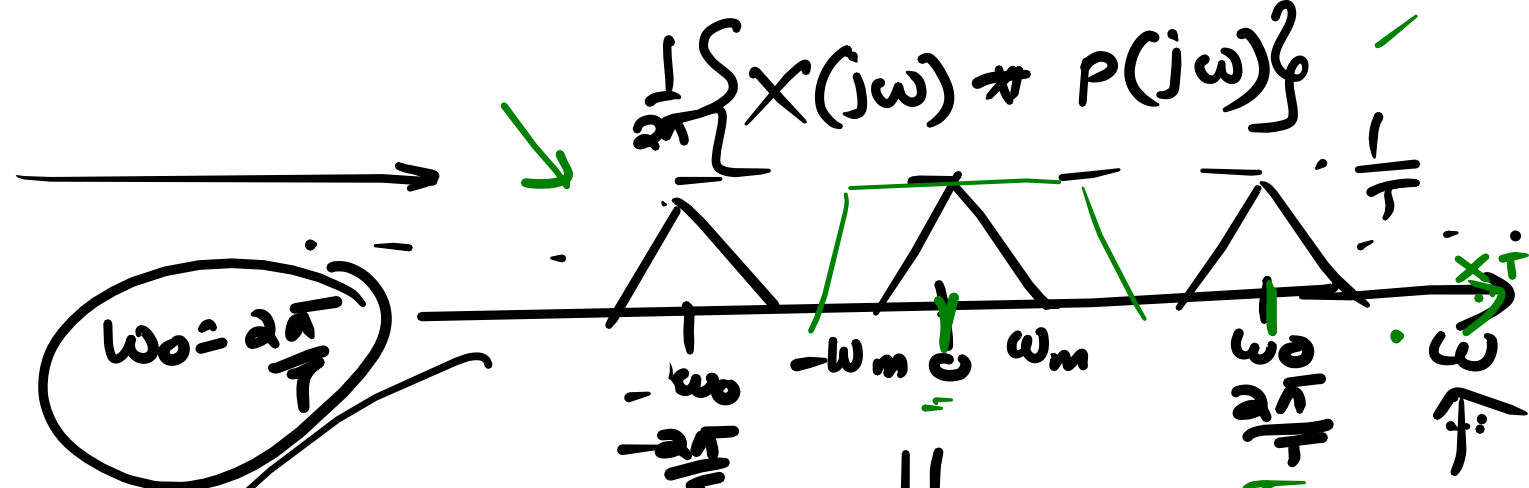


$$x(nT) \approx x[n]$$

Scale the frequency axis by T & correspondingly inversely scale in the time domain

⇓

move to discrete time domain



⇓

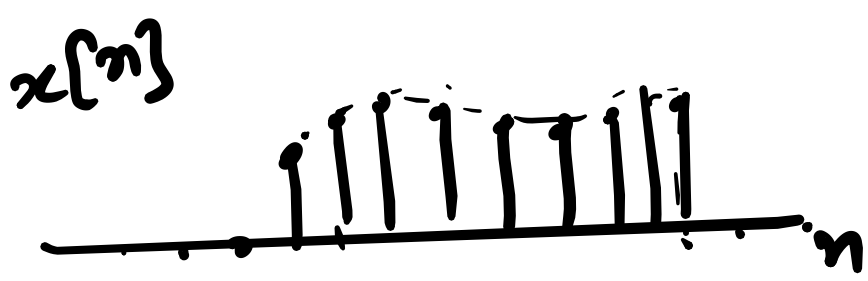
In the continuous time domain

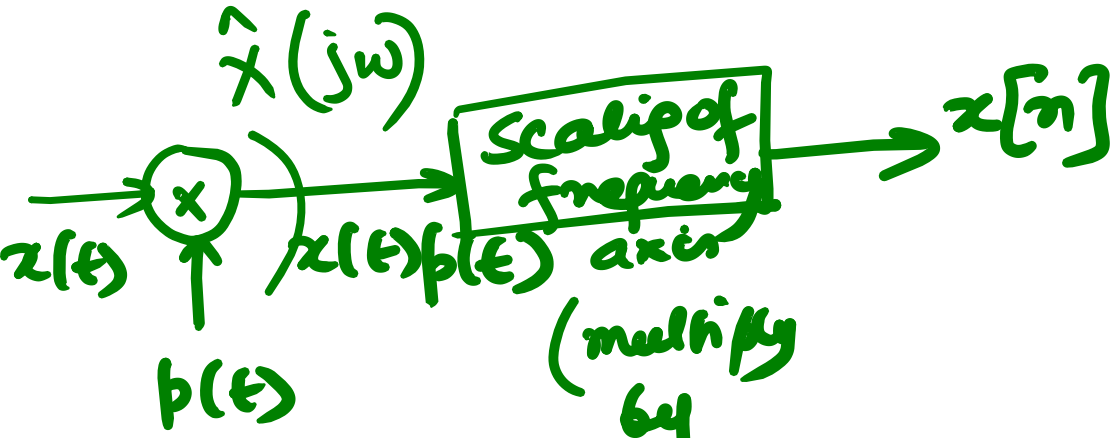
$\omega \equiv$ rad/s. (Units)

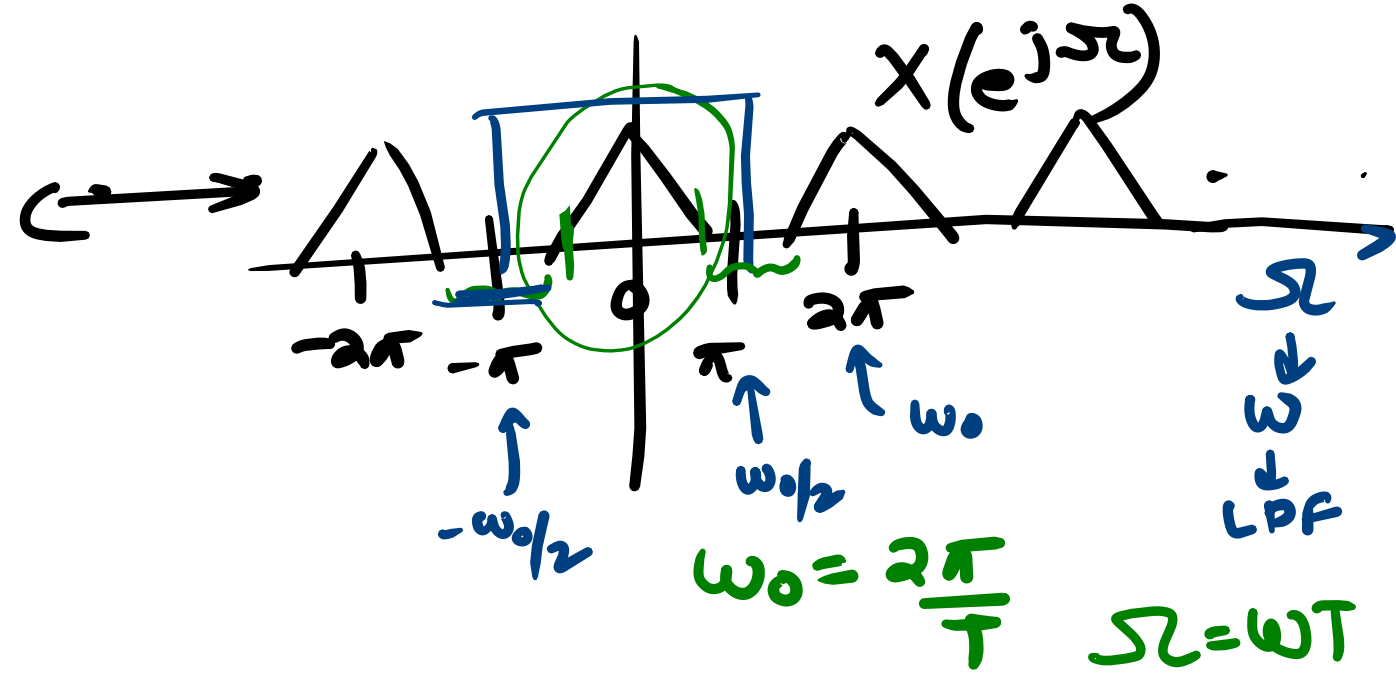
In the D.T. domain

$\Omega \equiv$ rad (Units)

$\Omega = \omega T$ | $\omega = \Omega / T$







$$\frac{1}{2\pi}\left[X(j\omega) * P(j\omega)\right] = \frac{1}{T}\sum_{k=-\infty}^{\infty} X\left(j(\omega - k\omega_s)\right) = \hat{X}(j\omega)$$

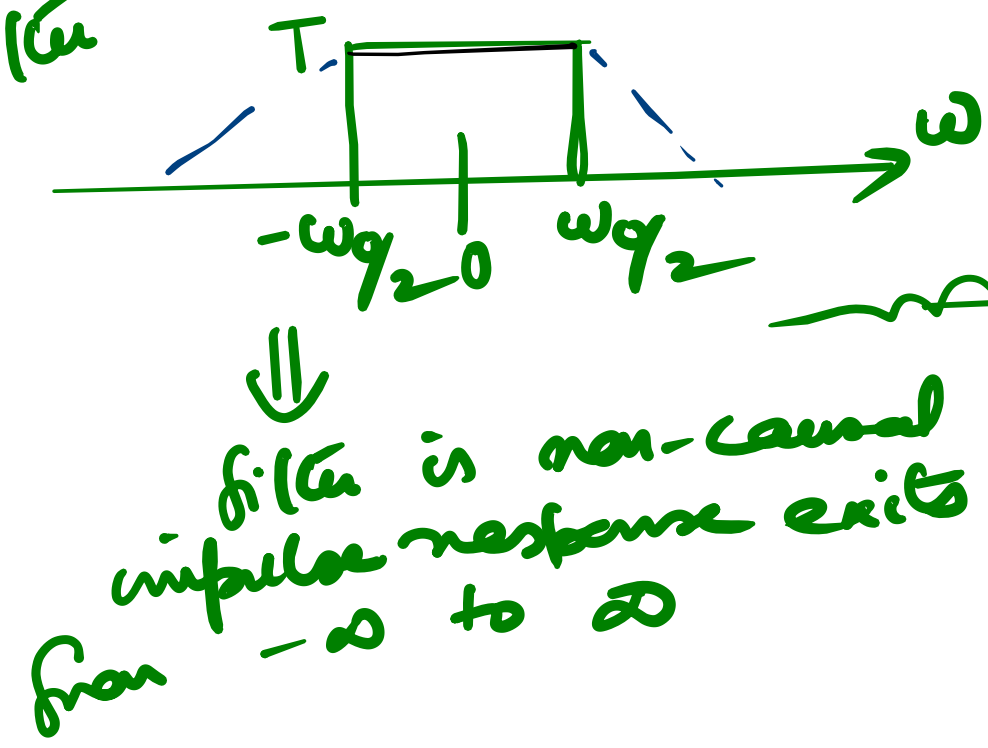
$$X(e^{j\Omega}) = \hat{X}\left(j\frac{\Omega}{T}\right) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X\left(e^{j\left(\frac{\Omega - k2\pi}{T}\right)}\right)$$

If we want to come back

from $x[n] \longrightarrow x(t)$

$\Omega \longrightarrow \omega \longrightarrow$ Use a lowpass filter in continuous time domain with bandwidth from $-\omega_q/2$ to $\omega_q/2$

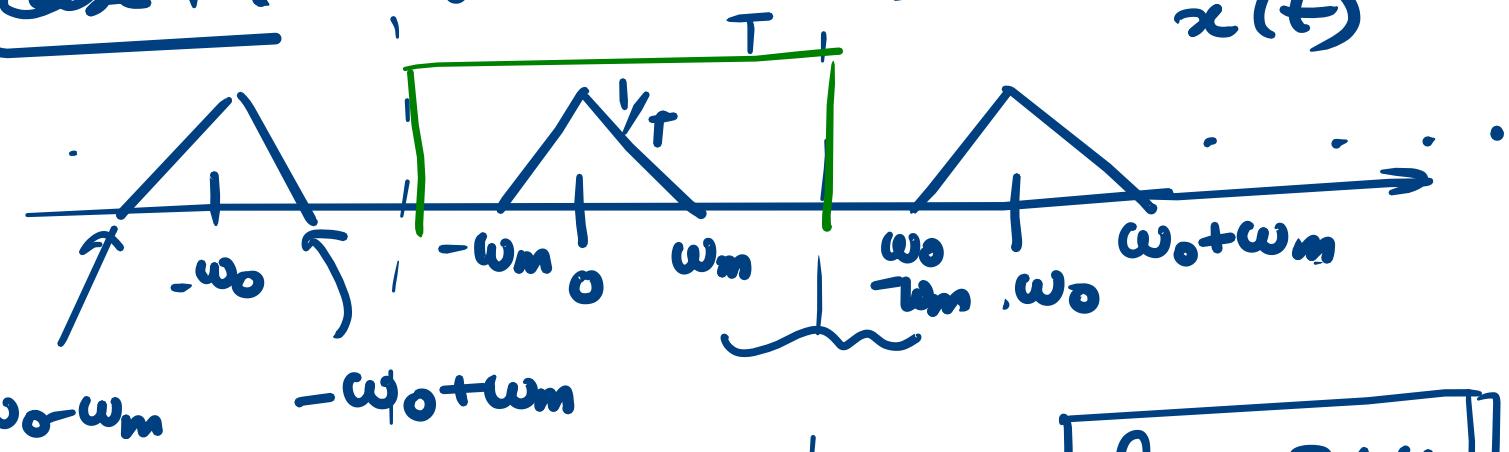
C.T. Ideal filter



filter is non-causal

impulse response exists from $-\infty$ to ∞

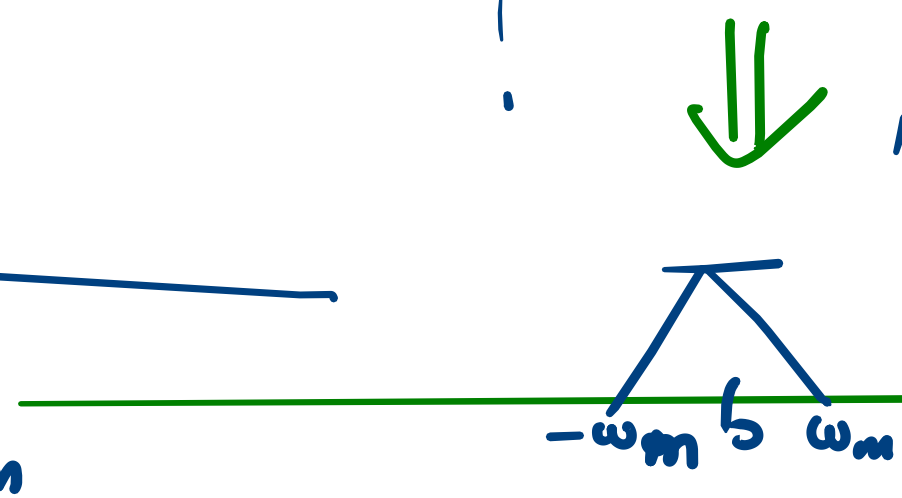
Oversampling → Case 1 : $\omega_0 - \omega_m > \omega_m \Rightarrow \omega_0 > 2\omega_m$



$x(t) \rightarrow \hat{x}(t) \rightarrow x[n]$

$\tilde{x}(t) \leftarrow \hat{x}(t) \leftarrow$

Goal $\boxed{\tilde{x}(t) = x(t)}$



$\omega_0 - \omega_m \gtrless \omega_m$

If $\omega_0 - \omega_m = \omega_m$

$f_m = 8\,KHz$

$f_0 > 16\,KHz$

$\omega_m = 2\pi f_m$

$\omega_0 = 2\pi f_0$

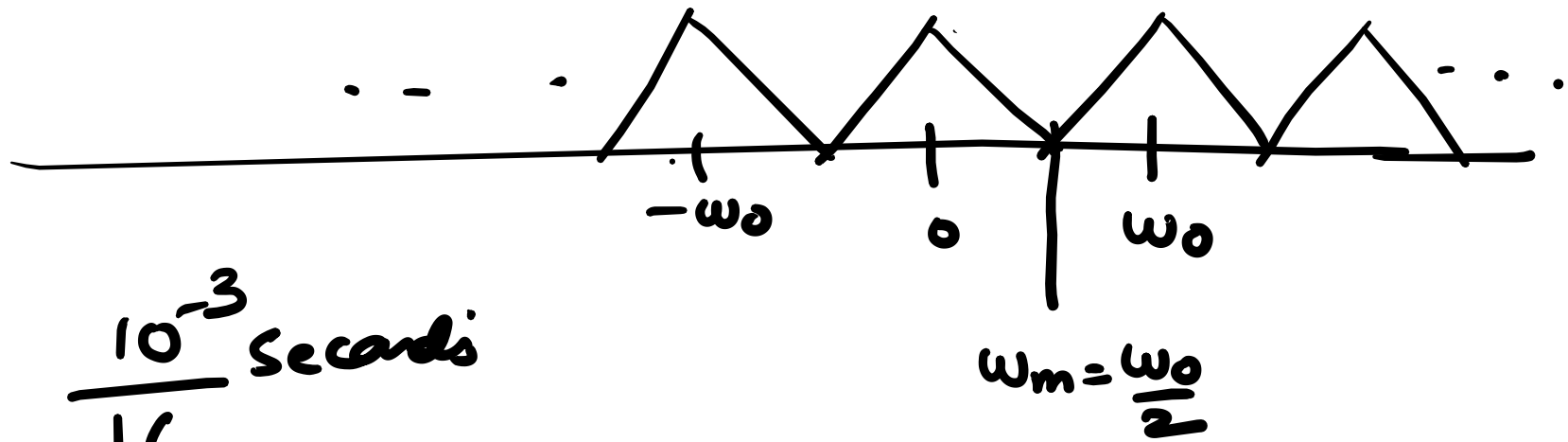
$T < \frac{10^{-3}}{16}\,s$

Case-2

$\omega_0 - \omega_m = \omega_m$

$\omega_0 = 2\omega_m$

At Ideal sampling rate



$\omega_0 = 2\omega_m$

Nyquist rate

$T = \frac{10^{-3}}{16}$ seconds

Requirement, Strictly an ideal Low pass filter

$\omega_0 \geq 2\omega_m$

Nyquist Sampling Theorem : Given a band limited signal $x(t)$, if we sample it at a rate/frequency $\omega_0 \geq 2\omega_m$, we can recover/reconstruct original signal.

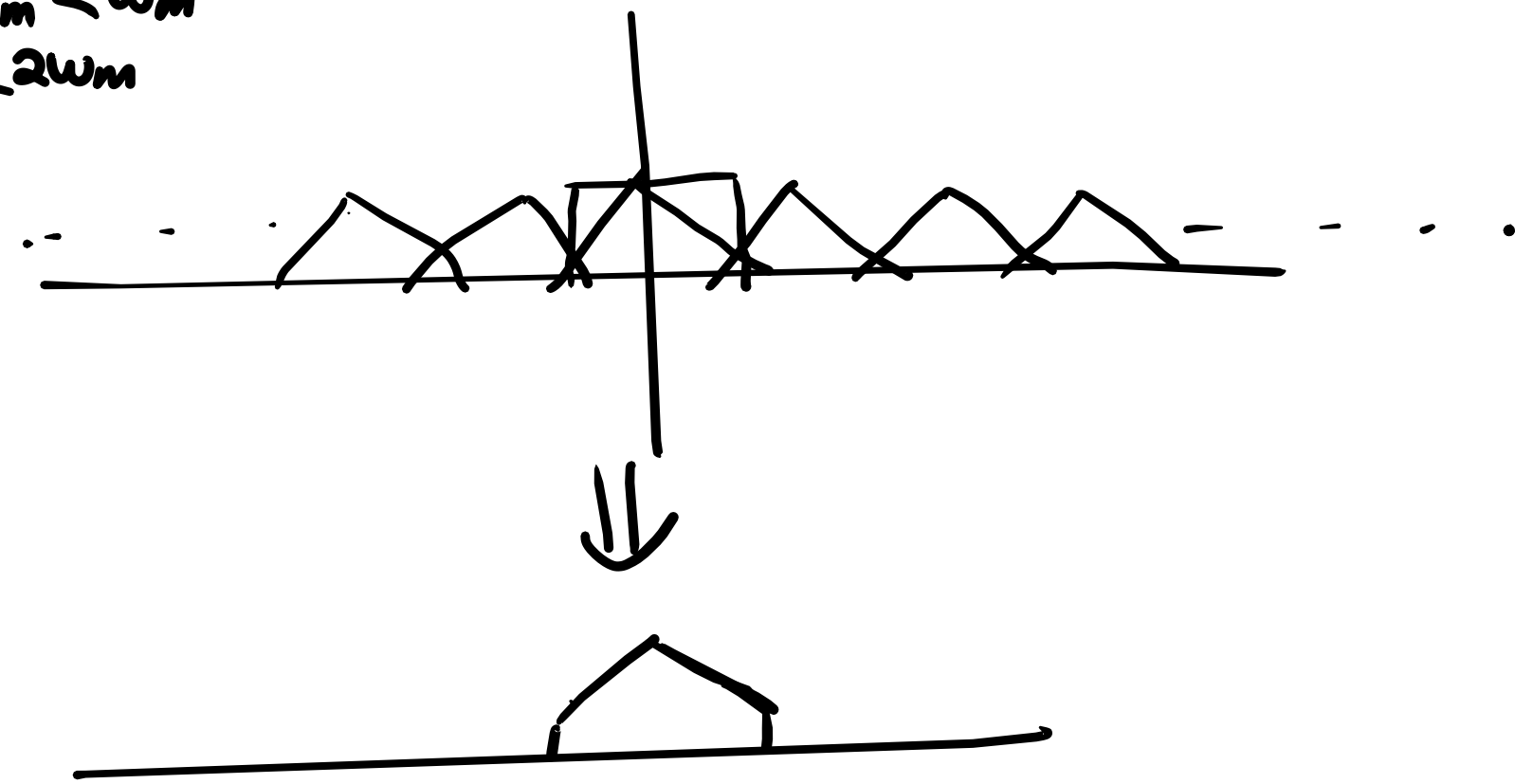
Case -3 : $\omega_0 - \omega_m < \omega_m$

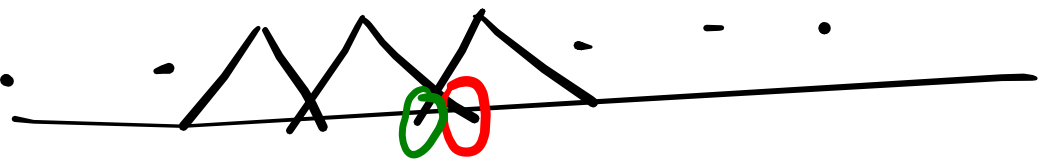
$\omega_0 < 2\omega_m$

Undersampling

Aliasing

$T > \frac{10^{-3}}{T_6}$





Example $x(t) = \cos \omega_m t$ $f_m = 8\,KHz$

Case 1: $\omega_0 > 2\omega_m$ $f_0 = 20\,KHz$

Case 2: $f_0 = 15\,KHz$

Case 3: $f_0 = 6\,KHz$

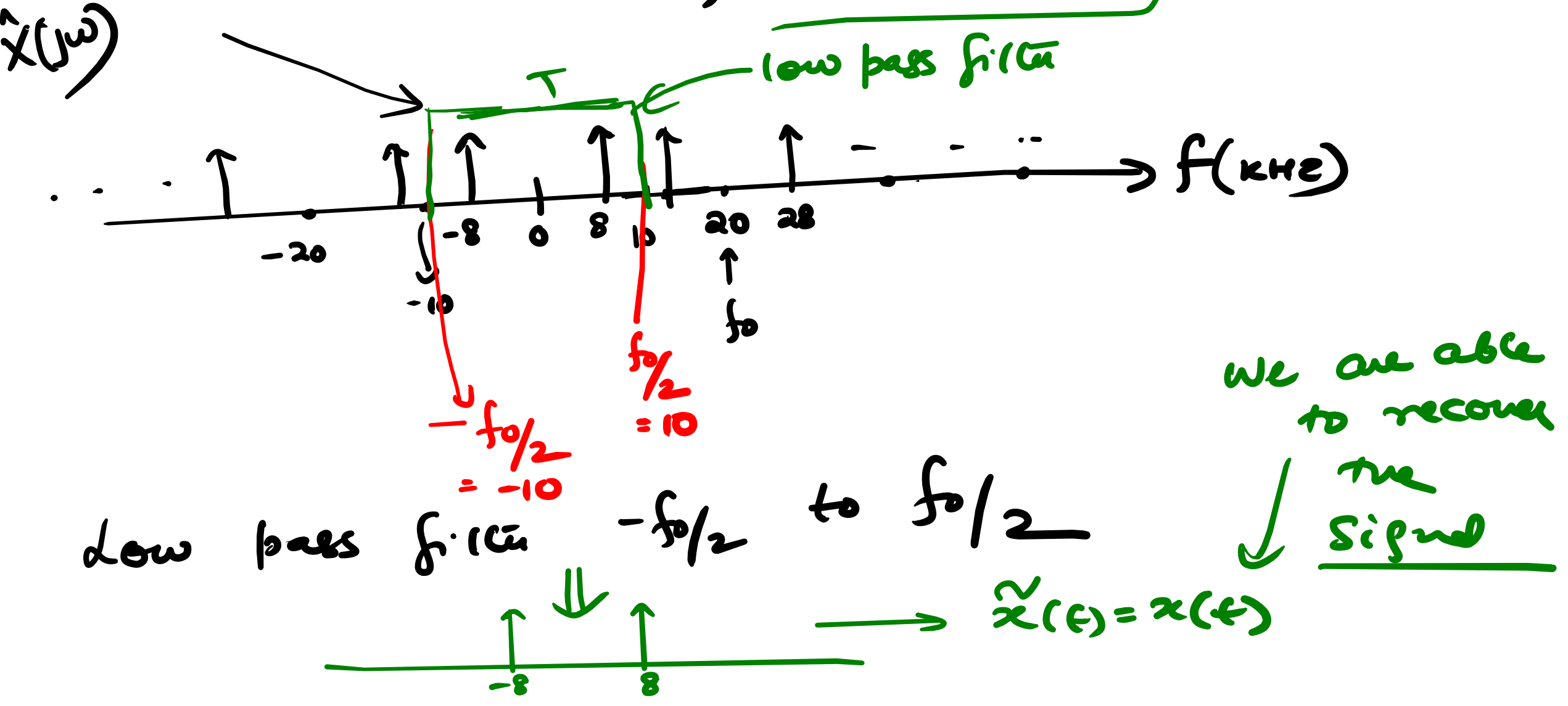
} Aliasing will happen

Case 1:

$x(t) = \cos \omega_m t$ $\quad f_m = 8$ kHz

$f_0 = 20$ kHz ← case of our sampling

$f_0 > 2 f_m$



We are able to recover the signal

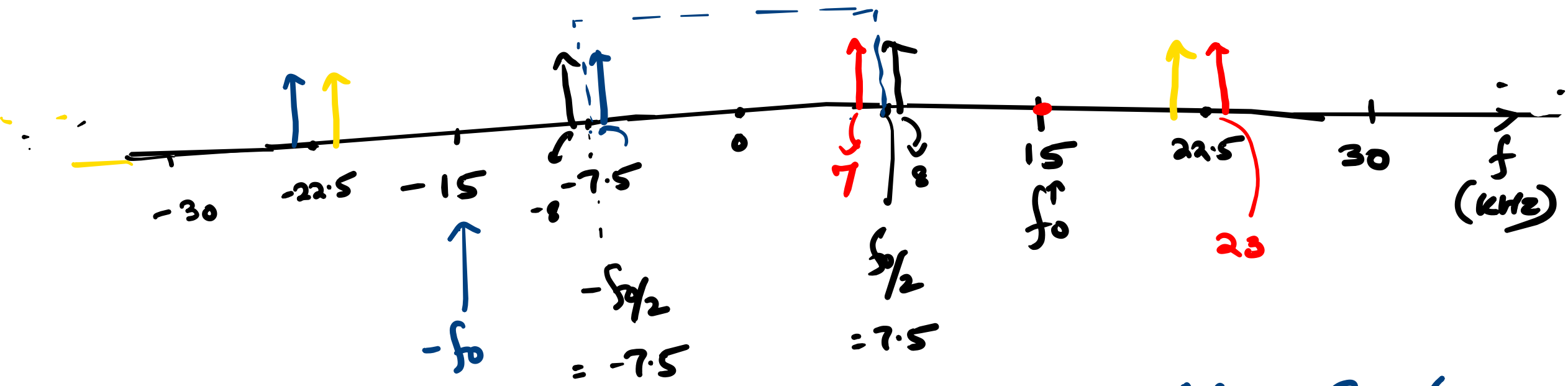
Low pass filter $-f_0/2$ to $f_0/2$

Case-2

$x(t) = \cos \omega_m t$

$\omega_m = 2\pi f_m$

$f_m = 8\,kHz$, $f_0 = 15\,kHz$



$-f_0/2 \longrightarrow f_0/2 \longrightarrow$ filter

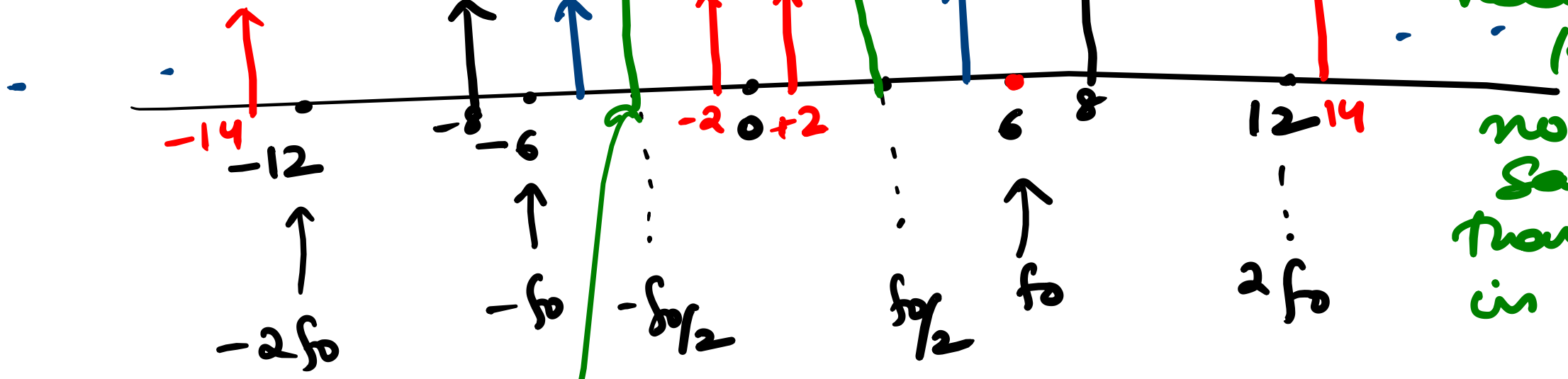
$\hat{x}(t) = \cos \omega t$

where

$\omega = 2\pi f = 2\pi(7\,kHz)$

Case -3

$f_0 = 6\ KHz$, $f_m = 8\ KHz$



Undersampling collected relatively less no. of samples than required in a given time

$T > T_{Nyq}$.

filter

Sampled 8 KHz signal → Reconstructed 2 KHz signal

$\tilde{x}(t) = \cos \omega t$

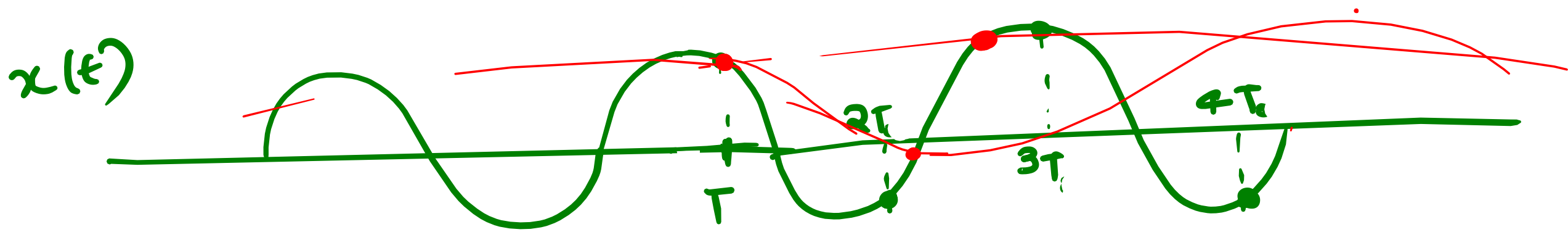
where $\omega = 2\pi f$

$f = 2\ KHz$

$x(t) = C_2(\omega_m t + \phi)$ $\omega_0 \geq 2\omega_m$

$\omega_0 = 2\omega_m \Rightarrow$ We are collecting 2 samples from one cycle of $x(t)$

$T = \frac{T_m}{2}$



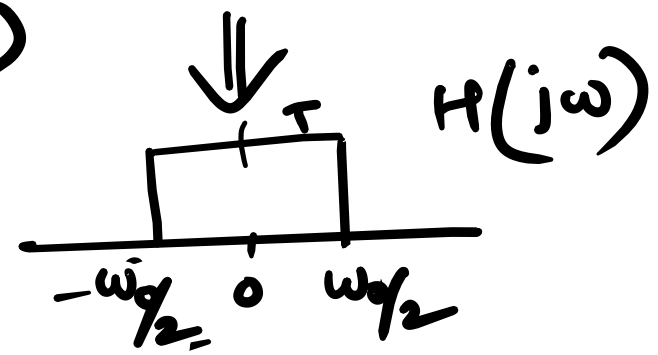
if $T_{Sampling} > \frac{T_m}{2}$

Reconstruction

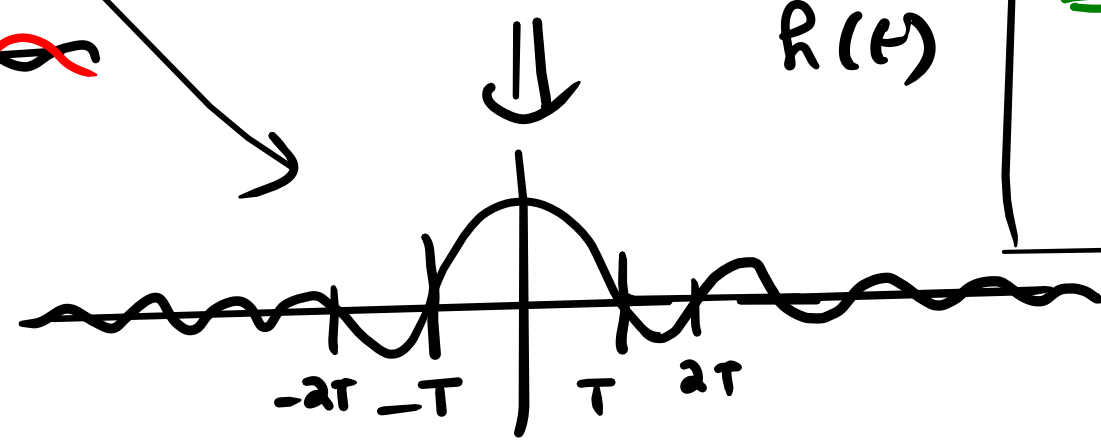
$\hat{x}(t)$ → [Low pass filter $R(t)$] → $\tilde{x}(t)$

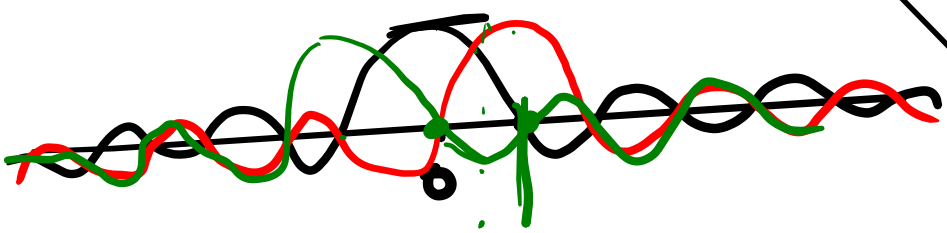
$$\hat{x}(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t-nT)$$

$$\hat{x}(t) * R(t)$$



$R(t)$





$x(0)$
$x(T)$
. .

$$\hat{x}(t) * h(t)$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t-nT) * h(t)$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\,h(t-nT)$$